

Supplementary Information

Protocol-Dependent Performance in Rotating Machinery Fault Diagnosis: Evidence from Sealed Compound Generalization

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S1. Evidence tiers and access states

The Sealed Compound Generalization Protocol (SCGP) is a prospective evaluation contract. The empirical evidence in the manuscript is labeled according to the strongest retained access history that it supports. The HANOI HUST compound track is a historical held-out confirmation under a documented local freeze. The three-code Paderborn pilot is a one-shot external scoring run, and the disjoint 12-code extension was frozen locally before its numeric archives were accessed.

Table S1: Access-state contract used in the manuscript.

State	Partition and access	Evaluation unit	Role
P0	Window-random; labels open	Window	Descriptive leakage diagnostic; excluded from primary inference
P1	Record-grouped; labels open	Record or bearing-aggregated endpoint	Loose primary reference; records from one bearing may cross partitions
P2	Bearing-grouped; labels open	Physical bearing	Leakage-controlled source comparison
P3	Nested bearing-grouped selection	Physical bearing	Selection-adjusted source estimate
P4	Confirmation access sealed by protocol	Physical bearing or declared external unit	Prospective confirmation state

The source-only information budget, denoted I_0 , permits HANOI source records and registered source metadata during model fitting and selection. It excludes numeric values and labels from the HANOI compound partition and from external boundary sets.

S2. Primary G2 partition–aggregation audit

The first two rows of Table S2 form the fixed-reference 2×2 comparison: partitioning varies between record-grouped and bearing-grouped, and the endpoint varies between record level and bearing aggregated. The third row adds nested selection within bearing-grouped outer folds.

Across the 100 nested outer folds, logistic–envelope configurations were selected 54 times, extra-trees–envelope configurations 29 times, and RBF-SVM–envelope configurations 17 times. The

Table S2: Primary G2 repeated-split means across 100 registered splits.

Protocol cell	Record AUROC	Record exact	Bearing AUROC	Bearing exact
Record-grouped, fixed	0.951	0.776	0.959	0.791
Bearing-grouped, fixed	0.828	0.510	0.860	0.490
Bearing-grouped, nested	0.777	0.420	0.803	0.406

split-level results are correlated resamples of the same 19 physical bearings and are not treated as independent experiments.

S3. Secondary G3 model-family audit

All methods in Table S3 use the same 100 bearing-grouped outer splits, the I0 source-only information budget, and physical-bearing aggregation. This secondary audit leaves the primary model choice unchanged.

Table S3: Source-only model-family comparison.

Method	AUROC	AUPR	Balanced accuracy	Macro-F1	Exact set
Fixed logistic reference	0.862	0.835	0.742	0.704	0.490
Random forest	0.785	0.750	0.666	0.628	0.458
GroupDRO-style logistic	0.845	0.825	0.721	0.680	0.482
MiniROCKET	0.751	0.708	0.682	0.628	0.556
WDCNN-style	0.630	0.608	0.530	0.386	0.116
ResNet1D	0.609	0.607	0.521	0.373	0.120
InceptionTime-style	0.569	0.565	0.516	0.370	0.120

S4. Legacy NOACE comparator statistics

The legacy audit predates the matched G2/G3 analysis and uses a different candidate pool and endpoint. Its results are retained for provenance under their original access state.

Table S4: Legacy 19-bearing source comparison.

Setting	AUROC	Balanced accuracy	Exact set	Brier	Hamming
SCGP source reference	0.814	0.743	0.544	0.161	0.199
NOACE-classical	0.809	0.742	0.579	0.216	0.211

Comparator	Exact set	Difference (pp)	95% CI	Holm-adjusted p
NOACE-classical	0.579	-3.51	[-28.1, 21.1]	1.000
NOACE-physics	0.579	-3.51	[-28.1, 21.1]	1.000
NOACE-deep	0.544	0.00	[-17.5, 15.8]	1.000

Here, the difference is the source-reference minus comparator difference in bearing-level exact-set accuracy. Confidence intervals use 10,000 bearing-cluster bootstrap resamples. The inferential support consists of 19 physical bearings.

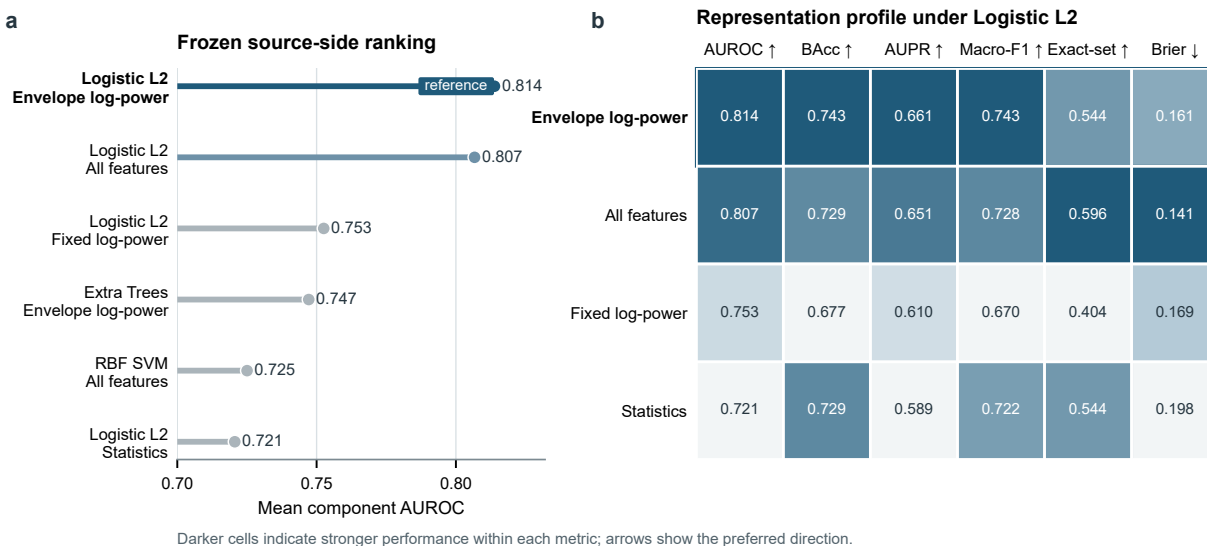


Figure S1: Legacy frozen source reference and representation profile. (a) Six model–representation pairs ranked by mean component AUROC on 19 source bearings. (b) Four representations compared with `logistic_l2`; darker cells denote better performance after reversing the Brier-score direction. `envelope_log-power` leads component discrimination, whereas `all` has higher exact-set accuracy and a lower Brier score. Values retain the endpoint and access state of the legacy audit.

S5. HANOI HUST held-out compound boundary

The held-out HANOI HUST track contains 14 compound bearings. The fixed logistic–envelope reference achieved mean component AUROC 0.811 and exact-set accuracy 2/14. NOACE-classical and the MLP comparator produced exact-set accuracy 0/14. Resampling 14 identical zero outcomes gives a degenerate empirical bootstrap interval of 0–0. A two-sided Clopper–Pearson sensitivity interval for 0/14 is 0–0.232, which better communicates population uncertainty beyond the observed boundary.

This track is described as a historical held-out confirmation under a documented local freeze. It is not described as a publicly time-stamped prospective lockbox.

S6. Supplementary heterogeneous boundary checks

The Mendeley Pump and Zenodo 14388093 analyses use different tasks, algorithms, and evaluation units from HANOI HUST. They are retained as supplementary checks and are not treated as replications of the HANOI classifier.

S7. Paderborn three-code pilot registry

The three-code pilot contains K001 (healthy), KA01 (outer-race damage), and KI01 (inner-race damage), with 80 records per code and 240 records in total. The source-trained logistic–envelope

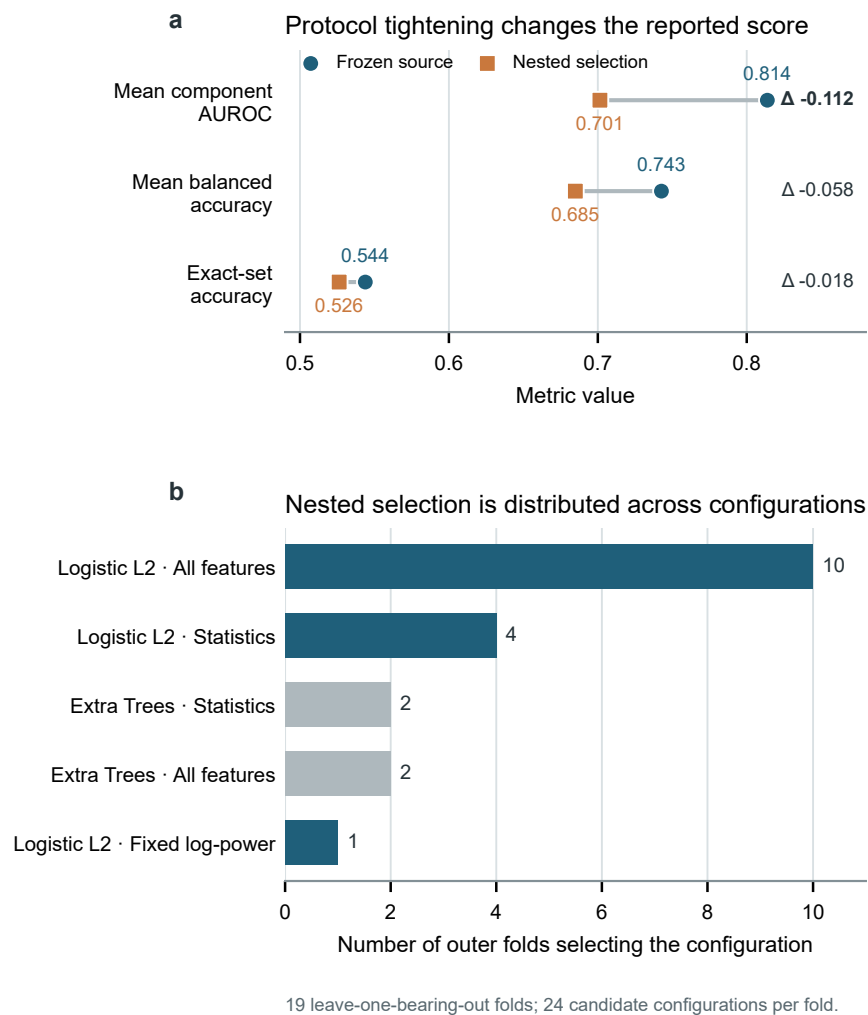


Figure S2: Legacy protocol tightening and selection stability. (a) Frozen source and nested physical-unit point estimates for mean component AUROC, balanced accuracy, and exact-set accuracy on 19 source bearings. Lines show state-specific changes; distinct metrics are not averaged. (b) Family-representation selections across 19 outer folds after pooling hyperparameters. Each fold screens 24 configurations. Counts are selection frequencies, not uncertainty estimates, and candidate pools differ across access states.

Table S5: Supplementary heterogeneous boundary results.

Resource	Evaluation unit	Recorded result and interpretation
Mendeley Pump	24 state workbooks	Source anchors were recovered for 9/9 states. Unseen exact recovery was 3/15 and generalized exact recovery was 12/24. Each workbook contributes one state-level external unit.
Zenodo 14388093	320 records from one rig at four speeds	The source-safe reference reached component balanced accuracy 1.000 and exact-set accuracy 1.000. Generalized values were 0.875 and 0.750. The track provides a single-rig external check.

model, fold-fitted scaling rule, and threshold 0.5 were fixed before external scoring. No Paderborn record or label was used for model selection or threshold tuning.

The three-code registry is independent of an earlier 15-code confirmatory registry. That earlier registry retained `numeric_access_authorized: false`, contained different bearing codes, and was never scored as that protocol. The pilot registry, archive hashes, scoring script, feature cache, result JSON, and unit rows are included in Supplementary Data S1.

At bearing-code level, the frozen reference recovered the healthy code but missed both damaged codes. Inner- and outer-race AUROC were 0.000, and exact-set accuracy was 1/3. This three-code result provides initial negative cross-rig evidence at the bearing-code endpoint.

S8. Disjoint 12-code Paderborn extension

To increase independent-unit support, a new registry was frozen before numeric access. It selected four healthy codes (K002–K005), four outer-race codes (KA04, KA15, KA16, and KA22), and four inner-race codes (KI04, KI14, KI16, and KI18). All 12 codes are disjoint from K001, KA01, and KI01. The HANOI-only scaler, logistic-envelope model, $C = 10$, threshold 0.5, record-to-code averaging rule, and uncertainty procedure were fixed before the official archives were downloaded. Paderborn labels were not used for model fitting, representation choice, or threshold tuning.

The 12 official RAR files matched their HTTP content lengths and yielded 80 MAT records each. Archive hashes, the registry hash, execution-script hash, derived feature-cache hash, and 960-record target mapping were retained. An independent audit rebuilt all code probabilities and headline metrics from the frozen source and external feature caches.

Inner-race AUROC was 0.344 (stratified bearing-code bootstrap 95% CI, 0–0.688), and outer-race AUROC was 0.719 (0.250–1.000). Exact-set accuracy was 3/12, or 0.250 (95% Clopper–Pearson CI, 0.055–0.572). Exact recovery by state was 2/4 (healthy), 0/4 (outer race), and 1/4 (inner race).

This extension increases the independent external support to four codes per state. Its one-rig design and absence of a Paderborn ball-fault state define the scope of the result, and its access record remains distinct from the pilot.

Table S6: Bearing-code outcomes for the pre-access-frozen Paderborn extension. Probabilities are ordered as inner race, outer race, and ball; targets and predictions use the same order.

Code	Target	p_{inner}	p_{outer}	p_{ball}	Prediction / exact
K002	000	0.284	0.005	0.000	000 / yes
K003	000	0.245	0.017	0.000	000 / yes
K004	000	0.728	0.004	0.000	100 / no
K005	000	0.628	0.004	0.000	100 / no
KA04	010	0.515	0.242	0.013	100 / no
KA15	010	0.400	0.029	0.000	000 / no
KA16	010	0.348	0.412	0.000	000 / no
KA22	010	0.528	0.001	0.000	100 / no
KI04	100	0.347	0.026	0.012	000 / no
KI14	100	0.587	0.021	0.000	100 / yes
KI16	100	0.287	0.016	0.002	000 / no
KI18	100	0.268	0.059	0.000	000 / no

S9. Reproduction entry points

Raw HANOI HUST and Paderborn waveform archives are not redistributed. Supplementary Data S1 contains derived caches, predictions, manifests, audit records, and source code sufficient to inspect and reconstruct the reported tables from the retained derived data.

Purpose	Entry point
Primary G2 execution	<code>src/run_hanoi_hust_g2_protocol.py</code>
Primary G2 audit	<code>src/audit_hanoi_hust_g2_main_table.py</code>
Factorial endpoint reconstruction	<code>results/g2/factorial_endpoint_metrics.json</code> and record-prediction arrays
G3 model-family runs	<code>src/run_hanoi_hust_g3_*.py</code>
Legacy NOACE ablation	<code>src/run_hanoi_hust_noace_ablation.py</code>
Paderborn boundary run	<code>src/run_hanoi_hust_paderborn_external_lockbox.py</code>
Paderborn 12-code extension	<code>src/run_hanoi_hust_paderborn_twelve_code_</code> <code>extension.py</code>
Paderborn extension audit	<code>src/audit_paderborn_twelve_code_extension.py</code>
Physical-unit manifests	<code>results/g2/split_manifests/</code>
Feature caches	<code>artifacts/hanoi_hust/</code>